BOHOL EXPLORER
Tourist information & travel guide

PHONE USER GUIDE

Use Bohol Explorer on your cellphone

Step-by-step instructions for opening the app, signing in, finding destinations, using the map, downloading offline maps, and testing the guide without internet.

Version 1.0 · November 2026 · Works offline once saved on your phone

START HERE

Quick start in five steps

Use this if the XAMPP database and the API are already installed on your computer. Detailed instructions follow below.

1. **Connect your phone and computer to the same Wi-Fi.**
   Do not use mobile data for the first setup.

2. **Start Apache and MySQL in XAMPP.**
   On the computer, open the XAMPP Control Panel and press *Start* for both.

3. **Build and serve the app for your network.**
   Replace `192.168.1.25` with your computer's Wi-Fi IPv4 address (see Part 1).

   ```
   flutter build web --release --dart-define=API_BASE_URL=http://192.168.1.25/touris
   dart run tool/serve_web.dart 8123 --lan
   ```

4. **Open the printed address on your phone.**
   In Chrome or Safari, type the *Phone address* shown in the terminal, for example `http://192.168.1.25:8123`.

5. **Create an account, then download the offline map.**
   Sign in, open the cloud icon at the top, and tap *Download offline map*.

Important: a phone cannot use `localhost` to reach your computer. Always use the computer's Wi-Fi IPv4 address (for example `192.168.1.25`) in the phone browser and in the API address used to build the app.

PART 1

Prepare the computer and the phone

The website runs in the phone's browser, but the PHP API and MySQL database stay on your computer (or a hosted server).

1.1 What you need

- The project folder `tourist_guide_bohol_jerivee` on a Windows computer.
- Flutter installed and working (`flutter doctor` shows no blocking errors).
- XAMPP with Apache, MySQL, and PHP 8.1 or later.
- An Android phone or iPhone with Chrome or Safari.
- The phone and computer on the **same Wi-Fi network**.

1.2 Start the database and API

1. **Open XAMPP Control Panel** and start **Apache** and **MySQL**. Both status labels should turn green.

2. **Confirm the API responds on the computer.** Open this address in the computer's browser:

   ```
   http://localhost/tourist_guide_bohol_api/api/index.php?resource=destinations
   ```

   You should see JSON data with destination records. If not, see Troubleshooting.

3. **Make sure the new columns exist.** For a database created before the contact/activities/accommodation update, import `backend/sql/destination_guide_upgrade.sql` in phpMyAdmin, then import `backend/sql/database.sql` again. The import updates the nine sample destinations without deleting users.

1.3 Find the computer's Wi-Fi address

1. On Windows, press `Windows` + `R`, type `cmd`, and press Enter.

2. Run `ipconfig`.

3. Under **Wireless LAN adapter Wi-Fi**, find **IPv4 Address**. It usually looks like `192.168.1.25` or `192.168.0.12`.

4. Write it down. This is the address your phone will use. Do **not** use `127.0.0.1` or `localhost` — those mean "this device" and would point at the phone itself.

> **If the address changes:** home routers can assign a different IP after a restart. Check `ipconfig` again. If it changed, rebuild the app with the new API address (step 1.4).

1.4 Build the app with the correct API address

Stop any running server first with `Ctrl` + `C`, then run this in the project folder. Replace the IP with your own:

```
flutter build web --release --dart-define=API_BASE_URL=http://192.168.1.25/tourist_guide_bohol
```

This step is required. A build made with the default `localhost` API address will fail to sign in from the phone.

1.5 Serve the build to the phone

```
dart run tool/serve_web.dart 8123 --lan
```

The terminal prints something like:

```
Serving C:\tourist_guide_bohol_jerivee\build\web at http://localhost:8123
Phone address: http://192.168.1.25:8123
```

- Keep this terminal window open while using the app. Closing it stops the website.
- The `--lan` option is required for a phone. Without it, the server accepts connections only from the computer itself.
- If Windows Firewall asks, allow access on **Private networks**.

1.6 Test the connection on the phone

1. Open Chrome or Safari on the phone.
2. Type the **Phone address**, including `http://` and the port, for example `http://192.168.1.25:8123`.
3. Also test the API itself on the phone: `http://192.168.1.25/tourist_guide_bohol_api/api/index.php?resource=destinations`.
4. If the API page shows JSON, the phone can reach the backend. If it does not, fix that before trying to sign in.

PART 2

Install the guide on the phone

The app can be added to the home screen and opened like a normal app. It keeps its own icon and, on a secure address, can reopen offline.

Android (Chrome)

1. Open the app address in Chrome.
2. Tap the **three-dot menu** in the top-right corner.
3. Tap **Add to Home screen** or **Install app**.

4. Confirm with **Install** or **Add**. A Bohol Explorer icon appears on the home screen.

iPhone (Safari)

1. Open the app address in **Safari** (not inside Facebook or Messenger).
2. Tap the **Share** button (square with an arrow).
3. Scroll down and tap **Add to Home Screen**.
4. Tap **Add**, then open Bohol Explorer from the home screen.

> **Secure-address limit:** installable PWA features, the offline app shell, and phone location work only on **HTTPS** or **localhost**. A plain `http://192.168.x.x` address still runs the app and saves offline content, but Chrome may only create a shortcut, and a full page reload with no internet may not work. For a complete offline demonstration on Android, use the USB method in Part 6, or host the build on an HTTPS address.

PART 3

Create an account and sign in

Login is mandatory. Guests cannot browse the destinations, map, saved guides, or itineraries.

Create a new account

1. On the full-screen welcome page, tap **Create account** (the button on the photo panel, or the link under the form).
2. Enter your **name**, a valid **email address**, and a **password** of at least 8 characters.
3. Re-enter the same password in **Confirm password**.
4. Tap **Create account**. On success you are signed in and the Home page opens.

Sign in to an existing account

1. Enter the account's **email** and **password**.
2. Tap **Sign in**.
3. If the API is unreachable, a message explains that the PHP/MySQL API is unavailable. Start Apache and MySQL, or check the API address used at build time.

Stay signed in and sign out

- The app remembers a valid session. Reopening it later takes you straight to Home unless the token expired (12 hours) or the API rejects it.
- After one successful online sign-in, the cached profile also allows the guide to reopen while the API is offline.
- To sign out: tap the **profile/avatar** in the top-right corner, then **Sign out**.

> **No self-service password reset is built in.** Use the password you registered, or create a new account. Passwords are stored hashed by PHP; they are never stored as plain text by the app.

PART 4

Use the guide, step by step

On a phone, the bottom bar has **Home**, **Explore**, **Map**, **Saved**, and **Plans**.

4.1 Home page

- Shows a welcome banner, category shortcuts (Nature, Beaches, Culture, Caves), and popular destinations.
- Type in **Search destinations in Bohol…** and press Enter to search.
- Tap **View all** or **Explore all** to open the full destination list.
- Tap **Plan your Bohol trip** to open the recommended itineraries.

4.2 Search for a destination

1. Open **Explore** from the bottom bar.
2. Tap the search field with the hint **Search by destination, town, or keyword**.
3. Type a place, town, or activity — for example *Chocolate Hills*, *Panglao*, *snorkeling*, or *hotel*.
4. The number of results updates immediately. Tap the × in the field to clear the search.

4.3 Filter by category

1. On **Explore**, scroll to the row of category chips.
2. Tap a category: *All*, *Nature & Adventure*, *Beaches & Islands*, *Historical & Cultural*, *Caves & Waterfalls*, *Wildlife & Eco-Tourism*, or *Food & Local Experiences*.

3 Search and category filters work together. Tap **All** to remove the category filter.

4.4 Open a destination's travel information

Tap any destination card. The detail panel shows:

Entrance fee & operating hours
Current guidance; confirm locally because rates and schedules change.

Travel tip
Practical advice for the site, such as footwear, water, or wildlife rules.

Things to do
Activities available at or near the destination.

Where to stay
Accommodation areas and nearby towns, not a booking service.

Contact information
Official municipal tourism contacts and the provincial tourism office.

Map coordinates
Reference point for the visitor area, not a property boundary.

4.5 Save a destination and get directions

- **Favorite** — adds the place to *Saved → Favorites*.
- **Save offline** — keeps the written guide in *Saved → Offline* and in the Offline guide center. Tap again to remove it.
- **Mark visited** — adds it to *Saved → Visited* for your trip record.
- **Get directions** — opens Google Maps with the destination coordinates. You still need a connection for Google Maps.
- **View on Bohol map** — switches to the in-app map.

4.6 Use the map

1. Open **Map** from the bottom bar.
2. Pinch to zoom and drag to move. Red pins mark the destinations.
3. Tap a red pin (or a row under **Mapped destinations**) to open its guide.
4. Tap the **target icon** at the top-right of the map to show your approximate location.
5. Allow location access when the browser asks. If you deny it, the map still works, but your position is not shown.

Location requires HTTPS or localhost in modern browsers. On a plain LAN HTTP address, the browser may block it.

4.7 Recommended itineraries (Plans)

1. Open **Plans** in the bottom bar (labelled *Itineraries* on a wide screen).
2. Read the duration, number of stops, suggested route, and budget note.
3. Tap the **bookmark icon** to save a plan. A filled bookmark means it is saved.
4. Tap any stop to open its destination details.

Routes are examples, not confirmed tour packages. Check transport, fees, weather, and site access before travelling.

4.8 Saved library

1. Open **Saved** in the bottom bar.
2. Switch between the **Favorites**, **Offline**, and **Visited** tabs.
3. Tap **Open offline guide center** to manage downloaded content.

4.9 Account menu and administrator dashboard

- Tap the avatar in the top-right to see your name and email, and to sign out.
- Administrator accounts also see **Admin dashboard**.
- The dashboard has **Overview**, **All users**, **User activities**, and **Destinations** sections.
- In **Destinations**, tap **Add destination** to create a record, or edit an existing one. Fields include name, municipality, category, description, image URL, latitude, longitude, entrance fee, operating hours, travel tips, contact information, activities, and accommodations.

- After promoting an account to `admin` in phpMyAdmin, sign out and sign in again so the new role appears.

PART 5

Download content for offline use

Do this while the phone has a working connection and the computer's server is running. Downloading prepares the guide for a trip with weak or no mobile data.

5.1 Open the Offline guide center

- Tap the **cloud icon** at the top of the screen (cloud with a check when online).
- Or open **Saved** → **Open offline guide center**.
- The page shows **Cached places**, **Map tiles**, **Storage used**, and **Map saved**.

5.2 Download the offline map

1. Scroll to **Bohol offline map**.
2. Tap **Download offline map**.
3. Keep the app open. A progress bar shows *x of y tiles*. You can tap **Cancel** to stop.
4. Wait for the message **"Offline map saved. You can now open the map without the internet."**
5. Confirm that **Map tiles** is greater than 0 and the card shows a green **READY** badge.

The download covers Bohol overview zoom levels plus a closer area around each destination. It is intentionally limited for the OpenStreetMap tile usage policy; it is not a complete navigation dataset.

5.3 Save individual destination guides

1. Open a destination and tap **Save offline**.
2. The guide appears under **Saved** → **Offline** and in the **Saved for offline reading** list.
3. Open each destination once while online so its photo is cached too.
4. Tap **Save offline** again to remove it.

5.4 Update or remove downloaded content

- **Update cached information** — refreshes the written catalogue from the API.

- **Update offline map** — downloads tiles again after a map update.
- **Remove** — opens *Remove offline map?*. This deletes downloaded tiles and cached photos, but keeps the written guides and saved destinations.

5.5 What is stored where

CONTENT	STORED ON THE PHONE	AVAILABLE OFFLINE
Destination names, descriptions, fees, hours, tips, contacts, activities, accommodation guidance	IndexedDB catalogue	Yes, after the first successful API load
Destination photos	IndexedDB image cache	Yes, once viewed or saved offline
Map tiles	IndexedDB tile cache	Yes, for downloaded areas and zoom levels
Signed-in profile and token	Browser local storage	Yes, until the token expires
Favorites, visited places, saved itineraries	Browser preferences	Yes, on this device
Admin changes, fresh API data, activity logging	PHP/MySQL server	No — needs a connection

PART 6

Test offline access on the phone

Run this test after downloading the map. It proves the offline features for a demonstration or thesis evaluation.

1. **Stay signed in and open every destination you want cached** before going offline.
2. **Open the Offline guide center** and confirm the tile count is above zero.
3. **Turn on Airplane mode** on the phone (or turn off Wi-Fi and mobile data).
4. **Reload the page.** A full offline reload needs the app-shell service worker, which requires HTTPS or localhost. See the secure-address note below.

5. **Confirm the orange banner** says *"Offline mode — showing guides and maps saved on this device."*
6. **Search and filter** destinations; cached results should still appear.
7. **Open the map**; downloaded areas should still display tiles.
8. **Turn Airplane mode off.** The banner disappears and the app refreshes the catalogue. Tap **Retry** if it does not.

> **Reloading offline on a phone:** service workers only run on `https://` or `http://localhost`. On `http://192.168.x.x`, the app and IndexedDB work while the page is already open, but a cold reload with no internet may fail. Test the full offline reload on the computer's Chrome (localhost) or use one of the options below.

Option A — Android USB, full offline demo

This makes the phone treat the app as `localhost`, which allows the service worker and a true offline reload.

1. Install Android Platform Tools (`adb`) on the computer.
2. On the phone, enable **Developer options** → **USB debugging**, connect USB, and allow debugging.
3. Keep the guide server running, then run:

```
adb reverse tcp:8123 tcp:8123
```

4. On the phone, open `http://localhost:8123`.
5. Sign in online, download the map, then enable Airplane mode and reload. The app shell and cached content should open.

The API can stay on the LAN address; it is only needed while online. If you also reverse port 80, the API can use `http://localhost/...`, but rebuilding is then required.

Option B — HTTPS hosting

For a realistic phone deployment, host `build/web` on an HTTPS static host and make the PHP API available over HTTPS at a public address. Rebuild with that API URL. Then install the app and test

offline mode on the phone. A phone cannot reach a PHP server that only runs on `localhost` of your computer.

Option C — Desktop Chrome (easiest for evaluation)

Open `http://localhost:8123` in Chrome on the computer. Service workers and IndexedDB work on localhost. Use DevTools → **Network** → **Offline**, then reload to demonstrate the offline shell, cached guides, and downloaded tiles.

PART 7

Troubleshooting

PROBLEM	LIKELY CAUSE	WHAT TO DO
“This site can't be reached” on the phone	Server stopped, wrong IP, different Wi-Fi, or firewall	Keep the terminal running; confirm the *Phone address*; check that both devices use the same Wi-Fi; allow the app through Windows Firewall on Private networks; do not use `localhost` on the phone.
“PHP/MySQL API is unavailable” or sign-in fails	Apache or MySQL stopped, wrong API URL, or API unreachable from the phone	Start Apache and MySQL; open the API URL in the phone browser; rebuild with `--dart-define=API_BASE_URL=http://YOUR-IP/tourist_guide_bohol_api/api/index.php`; check that the project is copied to `C:\xampp\htdocs\tourist_guide_bohol_api`.
Login says the password is wrong	Account not registered, typo, or different database	Use **Create account** to register; passwords need at least 8 characters. Confirm phpMyAdmin shows the user in `bohol_tourism_db.users`.
Blank or white screen	First load still running, cached build is stale, or browser is too old	Wait for the loading ring; refresh; clear site data for the address; use a current Chrome, Edge, or Safari; rebuild after source changes.
Map is blank or grey	No connection, no downloaded tiles, or tile requests blocked	Check the connection; open the map online once; download the offline map; allow `tile.openstreetmap.org`; note that areas and zoom levels outside the download are not cached.
Map tiles are partly missing offline	Only the overview and destination blocks were downloaded	This is expected. Download again with a connection, or stay within the covered Bohol area and zoom levels.
“Location is unavailable”	Permission denied, or the page is not a secure context	Allow location for the site; use HTTPS or localhost. On iPhone, check Settings → Safari → Location. The map still works without your position.
Offline reload fails on the phone	Page served over plain LAN HTTP, so no service worker	Use desktop localhost, Android `adb reverse`, or an HTTPS host. IndexedDB content is still there; only the app shell cannot start without a network.

PROBLEM	LIKELY CAUSE	WHAT TO DO
“Install app” is missing	Not HTTPS, unsupported browser, or already installed	Use Chrome on Android or Safari on iPhone; use HTTPS for a real install. “Add to Home screen” may still create a shortcut.
Photos do not appear offline	The image was never viewed while online	Open each destination once online, or mark it **Save offline**. Wikimedia images need a connection the first time.
Offline information looks outdated	The cache was saved earlier	Reconnect and tap **Update cached information** in the Offline guide center. Fees, hours, and contacts can change.
Download stops or storage is full	Connection dropped or browser storage quota reached	Keep the app open, free browser storage, remove unused site data, and retry. Clearing site data deletes cached guides, photos, tiles, and the saved session.
Need to start over	Stale cache or test data	Clear the site's storage in the browser settings, then reload and sign in again. On Chrome: site info → Delete data.

PART 8

Demonstration checklist

Use this sequence to show that every objective works during a thesis defense or class demo.

Tourism information

- ✓ Open a destination and show the description, entrance fee, operating hours, travel tip, activities, accommodation guidance, and contact information.
- ✓ Tap **Get directions** to show the external map link.

Destination search and filtering

- ✓ In **Explore**, search for *snorkeling*, *hotel*, or a municipality.
- ✓ Select *Beaches & Islands* and show that search and category filters combine.

Map and tourist navigation

- ✓ Open **Map**, tap a red pin, and open the destination guide.
- ✓ Tap the target icon and allow location to show the approximate position.

Offline access and offline maps

- ✓ In the Offline guide center, tap **Download offline map** and wait for the READY badge and tile count.
- ✓ Turn on Airplane mode, reload, and show the orange offline banner and cached destinations.
- ✓ Open the map and show downloaded tiles; search still works without a connection.
- ✓ Turn the connection back on and show the automatic refresh.

Recommended itineraries

- ✓ Open **Plans**, save an itinerary with the bookmark icon, and open one of its stops.

Accounts and administration (optional)

- ✓ Register a new account and show that browsing requires sign-in.
- ✓ Sign in as an administrator and show Overview, All users, User activities, and Destinations.

Before you present

- ✓ Charge the phone and keep the computer's terminal open.
- ✓ Confirm the phone and computer are on the same Wi-Fi and the API URL is the current LAN IP.

- ✓ Sign in and download the offline map in advance; cached content survives closing the browser.
- ✓ Have a second account ready in case the demo account's session expires.
- ✓ Keep this guide on the phone; it needs no internet.

> **Good to know:** signed-in actions (sign-in, searches, views, favorites, offline saves, visited places, itinerary saves, and admin changes) are recorded for system administration and evaluation. The app does not collect passwords, password hashes, or tokens for that activity view.

Data credits and accuracy

- Map data © OpenStreetMap contributors, available under the Open Database License. The offline download is limited and is not turn-by-turn navigation.
- Destination photos come from Wikimedia Commons; see `DESTINATION_IMAGE_CREDITS.md` in the project.
- Contact details come from official Bohol tourism pages. Fees, schedules, contacts, and access rules can change — confirm them before travel or publication.
- Map pins mark visitor areas, not property boundaries. Verify routes and coordinates on site.

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This file is self-contained. Once downloaded, it opens without an internet connection.